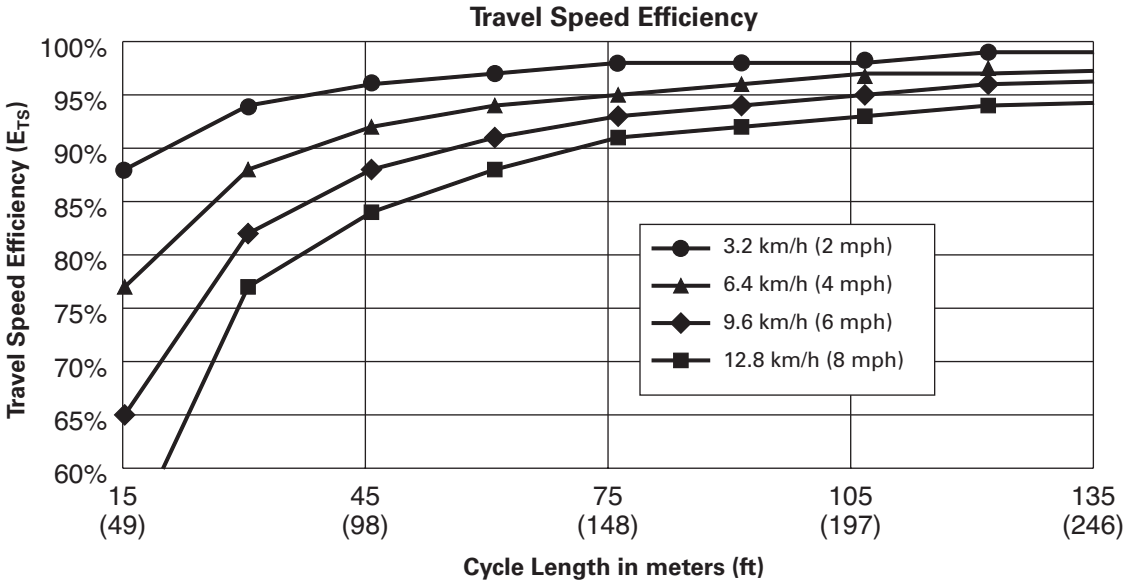
The tables in this section give production estimates for the following assumed conditions:

Nominal machine travel speed: 6.4 km/h (4.0 mph)
Overlap of rolling width: 15.2 cm (6.0 inches)

Table values give **representative** production rates for three common construction conditions: trenches, roads, and wide areas (> 15 m, or 50 ft).

Model	Drum Width cm	Drum Width in	Lift Thickness cm	Lift Thickness in	Passes Required		Production Estimates 3.7 m (12 ft) Trench	Production Estimates 9.15 m (30 ft) Road Base	Production Estimates Wide Areas
CS423E, CS44, CS44B	167.6	**66**	10.2	**4**	4	m^3/hr	159	249	249
						yds^3/hr	**209**	**326**	**326**
CS533E, CS54, CS56, CS54B, CS56B, CS64B, CS66B	213.4	**84**	15.2	**6**	6	m^3/hr	239	299	324
						yds^3/hr	**313**	**391**	**424**
CS64, CS68B	213.4	**84**	15.2	**6**	5	m^3/hr	—	373	405
						yds^3/hr	**—**	**489**	**530**
CS74, CS74B	213.4	**84**	15.2	**6**	4	m^3/hr	—	448	486
						yds^3/hr	**—**	**587**	**636**
CS76, CS76B	213.4	**84**	15.2	**12**	6	m^3/hr	—	598	648
						yds^3/hr	**—**	**782**	**848**
CS76 XT, CS78B, CS79B	213.4	**84**	15.2	**12**	4	m^3/hr	—	896	972
						yds^3/hr	**—**	**1174**	**1272**
CP44, CP44B	167.6	**66**	15.2	**6**	6	m^3/hr	159	199	249
						yds^3/hr	**209**	**261**	**326**
CP533E, CP54, CP56, CP54B, CP56B	213.4	**84**	30.5	**12**	6	m^3/hr	478	478	647
						yds^3/hr	**626**	**626**	**847**
CP76, CP68B, CP74B	213.4	**84**	30.5	**12**	6	m^3/hr	—	598	648
						yds^3/hr	**—**	**782**	**848**



Adjusting the Production Estimate

If the assumed conditions are not close to the actual construction conditions, the production estimates should be corrected. The production estimate from the table can be adjusted for 'actual' construction conditions by applying adjustment factors:

Q (actual) = Q (assumed) × Fs × Ft × Fp

Where: Q (actual) = adjusted productivity
Q (assumed) = productivity from table based on assumed conditions
Fs = adjustment for machine speed
Ft = adjustment for layer thickness
Fp= adjustment for no. of passes

The adjustment factors are determined by comparing the 'actual' conditions to the 'assumed' ones:

Fs = actual speed/assumed speed
Ft = actual thickness/assumed thickness
Fp = assumed passes/actual passes

Metric example

Actual Conditions — An 9.15-meter (full road width) base aggregate job is being completed with a compacted thickness of 15 cm. A CS44B is being used, operating at 4.0 km/h, and making 6 passes to achieve the desired compaction. The roller is overlapping its passes 6 inches.

For a 9.15-meter road base the table gives a CS44B productivity of 249 m^3/hr. Since the speed, thickness, and passes are *different* from the assumed conditions, we should adjust this estimate:

	Assumed	Actual
Speed	6.4 km/h	4.0 km/h
Thickness	10.2 cm	15 cm
Passes	4 passes	6 passes

Fs = 4.0 km/h/6.4 km/h = 0.6
Ft = 15 cm/10.2 cm = 1.5
Fp = 4 passes/6 passes = 0.7

The estimated production is adjusted using these factors:

Q (actual) = 249 m^3/hr × 0.6 × 1.7 × 0.7
= 178 m^3/hr (233 yds^3/hr)

English example

Actual Conditions — An wide area commercial site development job is being compacted in lifts of 8 inches. A CP56B is being used, operating at 4.0 mph, and making 4 passes to achieve the target density.

First, the table gives a CP56B productivity of 847 yds^3/hr. Since lift thickness and passes required are *different* from the assumed conditions, we should adjust this estimate:

	Assumed	Actual
Speed	4.0 mph	4.0 mph
Thickness	12 inches	8 inches
Passes	6 passes	4 passes

Fs = no correction necessary
Ft = 8 inches/12 inches = 0.7
Fp = 6 passes/4 passes = 1.5

The estimated production is adjusted using these factors:

Q (actual) = 847 yds^3/hr × 0.7 × 1.5
= 890 yds^3/hr (680 m^3/hr)

Notes on Productivity:

- For jobs that are relatively narrow, especially road construction jobs, it is important to understand that certain widths of construction will be more productive than others for a given compactor. A productive construction width will make the most use of each side by side pass required by the compactor in order to cover the width.
- Production estimates should be adjusted further if the length of the compaction cycles are shorter than 75 m (250 ft). Refer to the Travel Speed Efficiency chart to determine efficiency E_{TS}. For example, a compactor traveling at 6.4 km/h (4 mph) operating at cycle lengths of 150 ft has an E_{TS} of 0.91. Multiply Q (actual) by E_{TS}.

The table in this section gives production estimates for the following assumed conditions:

Compacted Layer Thickness	51 mm	**2 in**
Maximum Vibratory Frequency*	—	
Impacts Spacing	33 impacts/m	**10 impacts/ft**
Passes per Machine Width	**2**	
Compacted Material Density	2483 kg/m³	**155 lb/ft³**
Overlap of Rolling Width	152 mm	**6 in**
Overhang at Lane Edge	76 mm	**3 in**
Efficiency	**75%**	

*Maximum vibratory frequency varies by machine, refer to table.

Table values give **representative** production rates for common construction widths. If the actual width falls between two assumed widths, use the higher number to estimate production. Minor adjustments can normally be made in the rolling method to reach this higher production: reduce overlap or overhang, increase speed, or increase the cycle time.

			PAVING WIDTH						
Model	**Vibration Frequency**	**Units**	1.8 m **6 ft**	2.4 m **8 ft**	3.0 m **10 ft**	3.7 m **12 ft**	4.3 m **14 ft**	4.9 m **16 ft**	5.5 m **18 ft**
CB1.7	57 Hz	Tonnes/hr	154.6	160.4	164.0	166.6	–	–	–
	3420 vpm	**tons/hr**	**170.4**	**176.7**	**180.7**	**183.5**	**–**	**–**	**–**
CB1.8	57 Hz	Tonnes/hr	154.6	2062.9	200.5	196.9	–	–	–
	3420 vpm	**tons/hr**	**170.4**	**227.2**	**220.9**	**216.8**	**–**	**–**	**–**
CB2.5, CB2.5GC	64 Hz	Tonnes/hr	173.5	231.4	224.9	221.0	218.0	249.2	243.0
	3840 vpm	**tons/hr**	**191.3**	**255.1**	**248.0**	**243.5**	**240.4**	**274.7**	**267.9**
CB2.7, CB2.7GC, CC2.7, and CC2.7GC	64 Hz	Tonnes/hr	243.0	231.4	289.3	270.0	257.6	294.5	280.4
	3840 vpm	**tons/hr**	**267.9**	**255.1**	**318.9**	**297.7**	**284.0**	**324.7**	**309.0**
CB2.9	64 Hz	Tonnes/hr	243.0	231.4	289.3	270.0	315.0	294.5	331.3
	3840 vpm	**tons/hr**	**267.9**	**255.1**	**318.9**	**297.7**	**347.2**	**324.7**	**365.2**
CB4.0, CC4.0	55 Hz	Tonnes/hr	208.8	198.9	248.6	232.0	270.7	253.1	284.7
	3300 vpm	**tons/hr**	**230.2**	**219.2**	**274.0**	**255.8**	**298.4**	**279.0**	**313.9**
CB4.4	55 Hz	Tonnes/hr	208.8	278.4	248.6	298.3	270.6	309.3	284.8
	3300 vpm	**tons/hr**	**230.2**	**306.9**	**274.1**	**328.9**	**298.3**	**341.0**	**313.9**
CB7	53 Hz	Tonnes/hr	201.2	268.3	239.6	287.5	260.8	298.1	274.4
	3200 vpm	**tons/hr**	**221.8**	**295.7**	**264.1**	**316.9**	**287.5**	**328.6**	**302.5**
CB10	63.3 Hz	Tonnes/hr	240.5	320.6	400.8	343.5	400.8	356.3	400.8
	3800 vpm	**tons/hr**	**265.1**	**353.4**	**441.8**	**378.6**	**441.8**	**392.7**	**441.8**
CB13	63.3 Hz	Tonnes/hr	400.8	320.6	400.8	480.9	400.8	458.0	515.3
	3800 vpm	**tons/hr**	**441.8**	**353.4**	**441.8**	**530.1**	**441.8**	**504.9**	**568.0**
CB15, CB16	63.3 Hz	Tonnes/hr	400.8	320.6	400.8	480.9	400.8	458.0	515.3
	3800 vpm	**tons/hr**	**441.8**	**353.4**	**441.8**	**530.1**	**441.8**	**504.9**	**568.0**

Example

Actual Conditions — A 3.7 m (12 ft) lane is being paved with a compacted asphalt thickness of 10 cm (4 in). A CB10 is operating with a frequency of 42 Hz (2520 vpm) and 46 impacts per meter (14 impacts per ft) and making 4 passes to achieve target density. The roller is overlapping its passes 15 cm (6 in) and is overhanging the edges by 7.5 cm (3 in).

First, the table gives a CB10 productivity of 343 Tonnes/hr (378.6 tons/hr) for a 3.7 m (12 ft) paving width. Since the actual vibratory speed, thickness, impact spacing and passes are different from the assumed conditions, the estimate should be adjusted.

	Assumed	Actual
Vibratory Speed	63.3 Hz **3800 vpm**	42 Hz **2520 vpm**
Impact Spacing	33 impacts/m **10 impacts/ft**	46 impacts/m **14 impacts/ft**
Thickness	5 cm **2 in**	10 cm **4 in**
Passes	**2**	**4**

Fs = 42 Hz (2520 vpm)/63.3 Hz (3800 vpm) = 0.66
Fi = 33 impacts/m (10 impacts/ft)/46 impacts/m (14 impacts/ft) = 0.71
Ft = 10 cm (4 in)/5 cm (2 in) = 2.0
Fp = 2 passes/4 passes = 0.5

The actual, or adjusted, production estimate can then be determined from the following:

Q (actual) = 343 Tonnes/hr (378.6 tons/hr)
× 0.66 × 0.71 × 2 × 0.5 =
161 Tonnes/hr (177.4 tons/hr)

Notes on Productivity:

- Higher speed usually results in lower density achieved per pass.
- Productivity on uphill slopes may be reduced.
- Tabulated production estimates assume that 1 pass is used for re-positioning the machine at the beginning of the next run.

The tables in this section give production estimates for the following assumed conditions:

	Hot Mix Asphalt		Soil and Aggregate		Cold In-Place Recycled Asphalt	
Compacted Layer Thickness	51 mm	**2 in**	152 mm	**6 in**	203 mm	**8 in**
Maximum Propelling Speed	8 km/h	**5 mph**	8 km/h	**5 mph**	4.8 km/h	**3 mph**
Passes per Machine Width	**4**		**4**		**6**	
Compacted Material Density	2486 kg/cm^3	**155 lb/ft^3**	2085 kg/cm^3	**130 lb/ft^3**	2246 kg/cm^3	**140 lb/ft^3**
Overlap of Rolling Width	152 mm	**6 in**	152 mm	**6 in**	152 mm	**6 in**
Overhang at Lane Edge	76 mm	**3 in**	76 mm	**3 in**	76 mm	**3 in**
Cycle Time (2 passes)	**120 seconds**		**120 seconds**		**120 seconds**	

Table values give **representative** production rates for common construction widths. If the actual width falls between two assumed widths, use the higher number to estimate production. Minor adjustments can normally be made in the rolling method to reach this higher production: reduce overlap or overhang, increase speed, or increase the cycle time.

Hot Mix Asphalt		**PAVING WIDTH**						
		1.8 m	2.4 m	3.0 m	3.7 m	4.3 m	4.9 m	5.5 m
Model	**Units**	**6 ft**	**8 ft**	**10 ft**	**12 ft**	**14 ft**	**16 ft**	**18 ft**
CW16	Tonnes/hr	195.2	260.2	325.3	270.2	315.3	275.5	310.0
	tons/hr	**215.1**	**286.8**	**358.6**	**297.9**	**347.5**	**303.7**	**341.7**
CW34	Tonnes/hr	323.2	239.4	299.3	359.1	419.0	331.5	372.9
	tons/hr	**356.2**	**263.9**	**329.9**	**395.9**	**461.8**	**365.4**	**411.1**
Soil and Aggregate								
CW16	Tonnes/hr	490.1	653.4	816.8	678.6	791.7	691.9	778.4
	tons/hr	**540.2**	**720.3**	**900.4**	**748.0**	**872.7**	**762.7**	**858.0**
CW34	Tonnes/hr	811.6	601.1	751.5	901.8	1052.1	832.4	936.5
	tons/hr	**894.6**	**662.7**	**828.4**	**994.0**	**1159.7**	**917.5**	**1032.3**
PS360C	Tonnes/hr	882.2	653.4	816.8	980.2	1143.5	904.8	1017.9
	tons/hr	**972.4**	**720.3**	**900.4**	**1080.4**	**1260.5**	**997.3**	**1122.0**
Cold In-Place Recycled Asphalt								
CW16	Tonnes/hr	288.0	384.0	480.0	394.1	459.8	399.4	449.3
	tons/hr	**317.5**	**423.3**	**529.2**	**434.5**	**506.9**	**440.3**	**495.3**
CW34	Tonnes/hr	492.1	353.3	441.6	530.0	618.3	483.5	543.9
	tons/hr	**542.4**	**389.5**	**486.9**	**584.2**	**681.5**	**533.0**	**600.0**

Example

Actual Conditions — A 7.3 m (23'11") (full road width) base aggregate job is being completed with a compacted thickness of 200 mm (8 in). A CW16 is being used, operating at 6.5 km/h (4 mph), and making 6 passes achieve the desired compaction. The roller is overlapping its passes 152 mm (6 in).

First, the table does not show production for 7.3 m (23'11") so use the greatest width on the table: 5.5 m (18'1"). The table gives a CW16 productivity of 778.4 Tonnes/hr (858.0 tons/hr) for this paving width. We can expect that the actual productivity for 7.3 m (23'11") will be slightly higher than that. Since the speed, thickness, and passes are *different* from the assumed conditions, we should adjust this estimate:

	Assumed		Actual	
Speed	8 km/h	**5 mph**	6.5 km/h	**4 mph**
Thickness	152 mm	**6 in**	200 mm	**8 in**
Passes	**4**		**6**	

Fs = 6.5 km/h/8 km/h (4 mph/5 mph) = 0.8
Ft = 200 mm/152 mm (8 in/6 in) = 1.3
Fp = 4 passes/6 passes = 0.7

The estimated production is adjusted using these factors:

Q (actual) = 778.4 Tonnes/hr (858.0 tons/hr) × 0.8 × 1.3 × 0.7 = 567 Tonnes/hr (625 ton/hr)

Notes on Productivity:

- Ballast weight and tire pressure can significantly affect performance of a pneumatic tire compactor. Refer to machine specifications to choose the best configuration.
- Productivity on uphill grades and very thick layers (>127 mm, or 5 in) may be reduced due to a necessary reduction in speed.
- The 11-tire configuration for the CW16 is designed only for chip-and-seal applications. It is not recommended in other applications.

PIPELAYERS

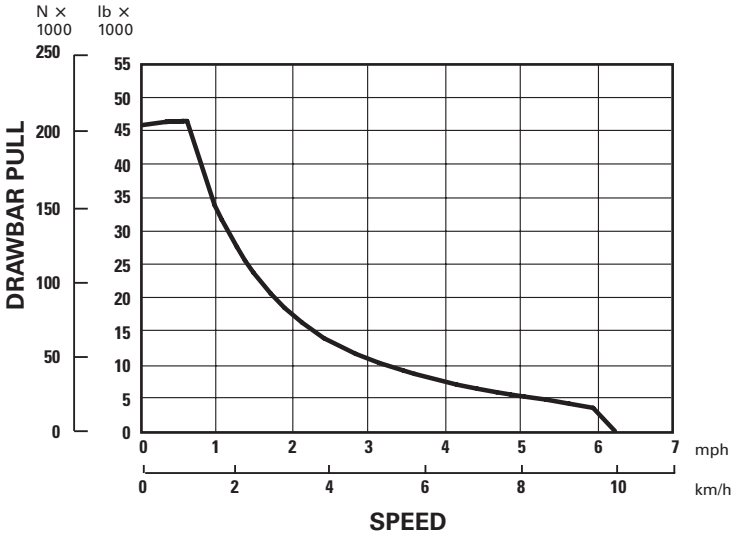
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Features:

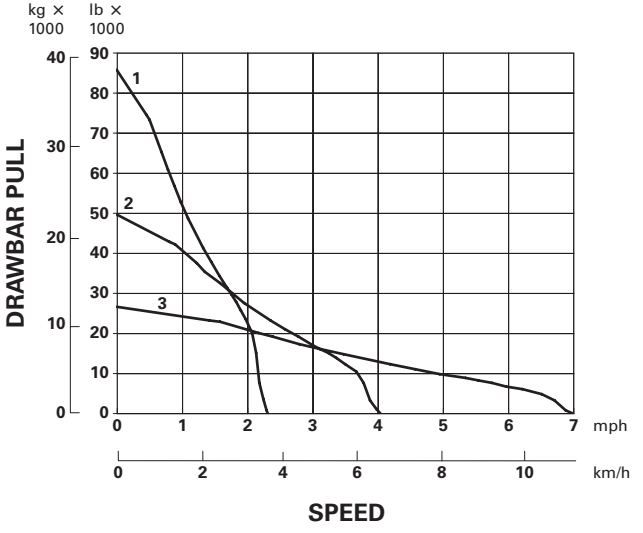
- **Hydrostatic transmission** on PL61.
- **Planetary power shift** transmission on PL72, PL83 and PL87 models.
- **Kick-out helps prevent boom bending** as boom approaches near-vertical.
- **Sealed and Lubricated Track.**
- **Simplified Controls** for all functions including raise, lower, quick-drop and power down, variable range and speed adjustments.
- **Modular design of major components and accessory drive system** for simplified repair.
- **Separate, self-energizing brakes** for boom and hook winches.
- **Positive** track pin retention (PL83 and PL87).
- **Hydraulic Drawworks** with two independently driven hydraulic motors for boom and hook winches.

PL61



NOTE: Usable pull will depend upon weight and traction of equipped tractor.

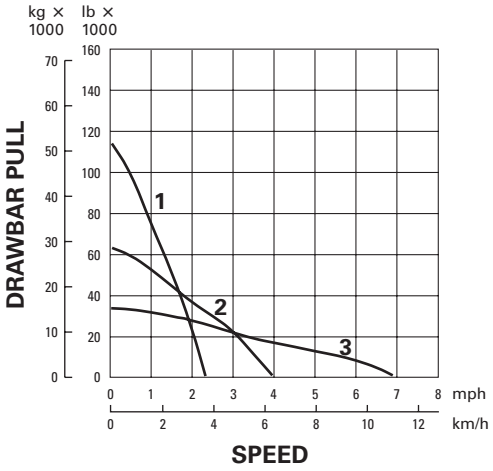
PL72



KEY

1 — 1st Gear
2 — 2nd Gear
3 — 3rd Gear

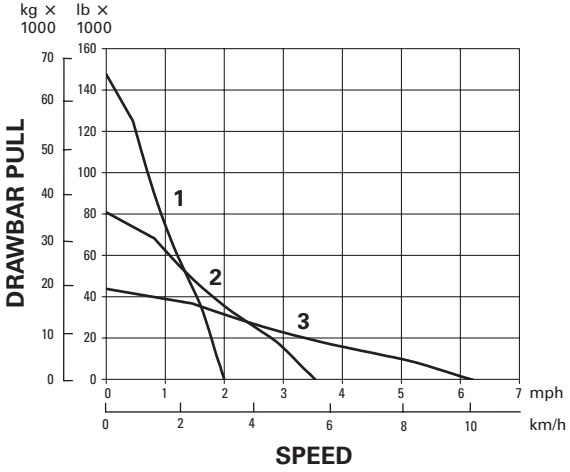
PL83



KEY

1 — 1st Gear
2 — 2nd Gear
3 — 3rd Gear

PL87



KEY

1 — 1st Gear
2 — 2nd Gear
3 — 3rd Gear

PIPELAYER APPLICATION CALCULATIONS:

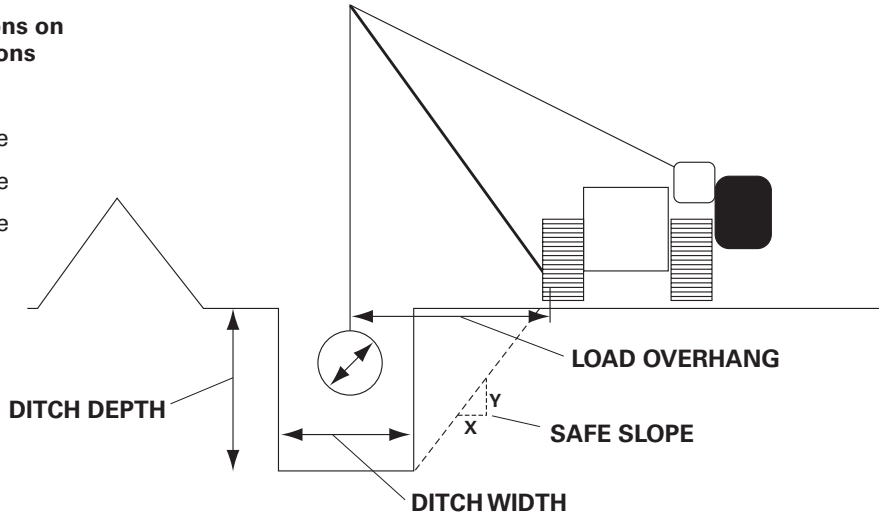
Typical pipelayer applications on flat, firm underfoot conditions

PL61 laying 8" to 16" pipe

PL72 laying 16" to 24" pipe

PL83 laying 24" to 36" pipe

PL87 laying 36" to 56" pipe



The chart above provides general information representing typical pipelayer applications. While the following scenario explores many of the variables involved in pipelaying it does not cover all the possible variables that must be considered by pipelaying contractors.

When sizing pipelayers for an application there are many considerations other than the machine's SAE rated lift capacity. These include but are not limited to:

- pipe diameter and weight per linear foot
- ditch width and depth
  ditch width is typically 2 × pipe diameter
  ditch depth is typically >2.5 × pipe diameter
- distance from the ditch (safe slope) required by soil stability conditions
  varies by local ground conditions but typically 2:1 (meaning the pipelayer must be 2 × ditch depth from the ditch edge)
- acceptable distance between pipe lifting points while suspended (to prevent bending)
  determined by the pipe's bending characteristics. If the lifting points are too far apart a pipe can sag enough due to its own weight that it will damage itself.
- the operating safety factor desired by the contractor
- the length of pipe that will need to be suspended while laying-in
  determined by pipe bending characteristics, terrain, etc.
- ground conditions, road bed preparation

An important consideration is the necessary load overhang. This is the distance from the center of the pipe to the tractor's left track rail. The load overhang required for an application can be estimated by:

- load overhang = safe slope × ditch depth + (0.5 × ditch width)

The pipelayer's rated load capacity at a specific load overhang (per ANSI/ASME B30.14) can be found in the load capacity graphs in this section of the performance handbook. Once the load capacity is determined the maximum lift point spacing can be estimated by:

- $\text{max lift point spacing} = \dfrac{\text{load capacity at load overhang}}{\text{safety factor} \times \text{pipe weight per linear foot}}$

The maximum distance between pipe lift points (based on pipe bending characteristics) may be a shorter distance than the maximum spacing between lift points as calculated based on pipelayer load capacity. If this is the case, then in order to avoid damaging the pipe, the shorter distance should be considered to be the maximum distance between pipelayers.

As an example, consider a project involving 0.5" wall 24" diameter pipe which has a weight per linear foot of 125.5 lb and the soil has a safe slope of 2. Using the above formulas:

- the ditch depth would be 3 × 2 ft = 6 ft deep
- the ditch width would be 2 × 2 ft = 4 ft
- the load overhang would be 2 × 6 ft + (0.5 × 4 ft) = 14 ft

Using the PL72's lift capacity chart we find that the PL72 has an ANSI rated load capacity of approximately 21,250 lb at a 14 ft load overhang.

When using rated load numbers it is important to understand that the lift capacity charts are based on SAE and ANSI test procedures that rate pipelayers on level, concrete surfaces. Working on softer underfoot conditions, working on slopes, (and other) can greatly reduce the pipelayer's load capacity.

If the contractor employs a safety factor of 2 then the maximum spacing between pipe lift points is:

$$\frac{21{,}250 \text{ lb}}{2 \times 125.5 \text{ lb/ft}} = 84.7 \text{ ft}$$

It is important to remember that this is the distance between the lift points, not the distance nose-to-tail between pipelayers. For this example, assume that 500 ft of pipe must be suspended during the laying-in process.

$$\frac{500 \text{ ft}}{84.7 \text{ ft per pipelayer}} = 5.9 \text{ which means that six pipelayers are needed}$$

The number of pipelayers required could also be determined by a second method:

$$\frac{\text{ft of pipe suspended} \times \text{pipe weight per ft} \times \text{safety factor}}{\text{rated load at overhang}}$$

In this case:

$$\frac{500 \text{ ft} \times 125.5 \text{ lb/ft} \times 2}{21{,}250 \text{ lb}} = 5.9 \text{ which again implies six pipelayers}$$

If, in this same example, soil conditions required a safe slope of 2.33 then the load overhang would have been 16 ft. At this load overhang the 90,000 lb lift pipelayer's rated load capacity is approximately 18,125 lb. Using the equations above, this results in 72.2 ft between lift points which means that seven 90,000 lb lift pipelayers are now necessary. Using the second method:

$$\frac{500 \text{ ft} \times 125.5 \text{ lb/ft} \times 2}{18{,}125 \text{ lb}} = 6.9 \text{ again implying that seven 90,000 lb lift pipelayers are needed}$$

Rather than adding another pipelayer, PL83's could be used. At a 16 ft load overhang the PL83 has a rated load capacity of 29,400 lb. This translates to 117.1 ft between lift points. If the pipe's bending characteristics will allow this space between lift points, the job could be done with only five PL83's.

EXTREME SLOPE OPERATION

The maximum fore and aft grade in static condition on which each track-type tractor or pipelayer will maintain proper lubrication is 45 degrees (100%). Consult Operation & Maintenance Manual (if applicable) for POWERTRAIN fluid level overfill requirements for operation on extreme slopes. Extreme slope operation is anytime the slope exceeds 25° (47%).

The ENGINE should never be overfilled with oil. This may lead to rapid overheating. For extreme slope operation, engine oil should be maintained at the full mark.

NOTE: Both ENGINE and POWERTRAIN fluid levels should be checked on level ground before working sidehills and slopes.

When working sidehills and slopes, consideration should be given to the following important points:

- Speed of travel — At higher speeds, inertia forces tend to make the tractor less stable.
- Roughness of terrain or surface — Ample allowance should be made where the terrain or surface is uneven.
- Mounted equipment — Bulldozers, sidebooms, winches, and other mounted equipment cause the tractor to balance differently.
- Nature of surface — New earthen fills may give way with the weight of the tractor. Rocky surfaces may promote side slipping of tractor.
- Track slippage due to excessive loads — This may cause downhill track to "dig in," increasing angle of tractor.
- Implements hitched to the drawbar — This may decrease weight on uphill track, e.g., logging arch, two-wheel wagon.
- Height of hitch on tractor — When a high drawbar is used the tractor is less stable than with the standard drawbar.
- Width of shoes — Wide track shoes tend to decrease "digging in", hence tractor is more stable.
- Operated equipment — Be aware of the stability and other performance features of the equipment operated by the tractor.
- Keep all attachments or pulled loads low to the ground for optimum stability.

NOTE: Safe operation on steep slopes may require special machine maintenance as well as excellent operator skill and proper equipment for the specific application. Consult Operation & Maintenance Manual (if applicable) for proper fluid level requirements.

SKID STEER LOADERS/ COMPACT TRACK LOADERS

CONTENTS

SKID STEER LOADERS/ COMPACT TRACK LOADERS

Features 12-1

Features:

- **The standard demand cooling fan** provides improved fuel efficiency and increased horsepower.
- **Industry leading sealed and pressurized cab option** provides a cleaner and quieter operating environment with excellent Work Tool visibility on D3 Series machines.
- **Ergonomically designed cab** provides maximum operator comfort and visibility.
- **Low-effort** joystick controls, armrest, and retractable seat belt for easy operation.
- **Available high-back, heated, air ride seat with seat mounted adjustable joystick controls and independently adjustable arm bars** make the D3 Series machines the industry leaders in operator comfort.
- **Deep skid resistant steps** make egress/ingress easy.

Features *(continued)*:

- **Hand and foot throttle** for continuous or variable engine speed.
- **Anti-stall feature/Electronic Torque Management** provide maximum rimpull and hydraulic power while preventing engine stalling.
- **Direct drive hystat pumps** eliminate drive belts for efficient power transfer.
- **Ground level access** to all daily service and routine maintenance points helps reduce machine downtime for greater productivity.
- **High performance power train** provides maximum performance and production capability on D and D2 Series machines through the Electronic Torque Management system, optional two speed travel and an electronic hand/foot throttle with decel pedal capability.
- **Cat® "Intelligent Leveling" system (ILEV)** provides industry leading technology and integration through available features such as dual direction self-level, work tool return to dig and work tool positioner.
- **Speed Sensitive Ride Control** option on D3 Series machines improves operation on rough terrain, enabling better load retention, increased productivity, and greater operator comfort.
- **Maximize machine capability and control with the Advanced Display** providing on-screen adjustment capability on D3 Series machines for implement response, hystat response, and creep control, multi-language functionality with customizable layouts, security system, and rearview camera.
- **High performance cooling systems and extended life coolant,** along with high performance engine and hydraulic oils, extend service intervals for low operating costs.
- **Deutsch electrical connectors with wires that** are color coded, numbered, and protected with nylon braiding.
- **Electro-depositioned** or "E" coat corrosion protection for long paint life.
- **High flow XPS hydraulics** are optional on most D3 Series models. The XE hydraulic package option on the 272D3 and 299D3 combines the high pressure of XPS with even higher flow for the most demanding hydromechanical work tool applications such as mulching, cold planing, and wheel saw cutting.
- **High flow hydraulics** on the 226D3, 232D3, 236D3, 239D3, 242D3, 249D3, 257D3, and 259D3 operate more demanding, rotating work tools such as brooms, cold planers, trenchers, landscape rakes, power rakes, tillers, snow blowers, brush cutters, and stump grinders.
- **Creep Control** allows the operator to select the optimal ground speed while maintaining maximum hydraulic flow to the attachment to achieve the best match between machine hydraulics, ground/operating conditions and attachment, especially effective with hydro-mechanical attachment.

SURFACE MINING EXTRACTION
Draglines
Electric Rope Shovels
Hydraulic Mining Shovels

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DRAGLINES

INTRODUCTION

Draglines are an important excavating tool used in many surface mining operations worldwide. These highly productive machines operate 24 hours a day, seven days a week and are able to reach depths of 79.8 m (262 ft) with capacities up to 116.2 m^3 (152 yd^3). Offering the lowest material removal cost per tonne (ton) and an average operating life of 40 years, draglines are the most productive and versatile machine in the industry. Caterpillar offers an extensive variety of dragline specifications and on-staff application engineers who will help determine the solution that best addresses specific needs.

With over 100 years of dragline experience and the largest operating fleet of draglines with buckets of 40 m^3 (52.3 yd^3) and above, coupled with an active installed base of over 300 machines, Cat draglines are the most efficient and proven overburden removal solution.

SAFETY AND SERVICEABILITY

Safety plays an integral role in everything we do at Caterpillar, from our factory floors to our service centers to our clients' mine sites. Caterpillar's commitment to safety is apparent in our product designs, which undergo risk assessments and are designed to meet strict codes and regulations. Stairways, walkways, platforms and access points are incorporated throughout Cat® draglines to ensure safe and convenient access for maintenance personnel. With safety and serviceability enhancing features incorporated into the operator cab, structures, surfaces and electrical equipment, among others, Cat draglines were designed with safety in mind.

Throughout this document, references to Tier 4 Interim/Stage IIIB/Japan 2011 (Tier 4 Interim) include U.S. EPA Tier 4 Interim, EU Stage IIIB, and Japan 2011 (Tier 4 Interim) equivalent emission standards. References to Tier 4 Final/Stage IV/Japan 2014 (Tier 4 Final) include U.S. EPA Tier 4 Final, EU Stage V, and Japan 2014 (Tier 4 Final) emission standards.

Throughout this document, references to Tier 1/Stage I include U.S. EPA Tier 1 and EU Stage I equivalent emission standards. References to Tier 2/Stage II/Japan 2001 (Tier 2) equivalent include U.S. EPA Tier 2, EU Stage II, and Japan 2001 (Tier 2) equivalent emission standards. References to Tier 3/Stage IIIA/Japan 2006 (Tier 3) equivalent include U.S. EPA Tier 3, EU Stage IIIA, and Japan 2006 (Tier 3) equivalent emission standards.

FEATURES

AC IGBT Electrics

Cat draglines are equipped with AC IGBT electrics which allow for greater machine uptime, lower operating costs, and faster cycle times, with AC providing up to 10% energy savings over the life of the machine.

- **Superior availability:** AC IGBT electric draglines have routinely demonstrated electrical availabilities of greater than 95%.
- **Greater reliability:** Fewer components including interchangeable inverters. IGBT systems require no fuses or circuit breakers.
- **Reduced maintenance:** No regular maintenance is required on IGBT power control modules. Motor maintenance on AC machines is reduced to greasing and replacing bearings every 30,000 hours.
- **Reduced inventory:** Hoist, drag, swing, and propel motions are all controlled by AC motors. Cat IGBT is a mine quality system with rugged welded cabinets to withstand harsh mining conditions.

Major Structures

Cat major structures are designed for extended performance in harsh mining conditions. Structures are manufactured using impact-resistant, high-strength steel with select welds of full-penetration, profiled and ground type. All structural welds undergo visual inspection, with critical welds also receiving MT, UT or X-ray inspection. Large furnaces are used to stress-relieve entire weldments for reduced susceptibility to cracking. Interiors of finished structures are painted white to facilitate field inspection.

Tri-Structure Design

The exclusive tri-structure design on Cat draglines reduces front end weight to enable optimization of boom configurations, allowing for increased load and reach. The tri-structure is manufactured with high impact strength steel for a simplified wide flange beam construction, and less maintenance points are required in comparison with an A-frame or mast assembly. Inventory stocking needs are reduced as the sheaves are interchangeable with both the fairlead and boom assemblies.

Cat Cab

Cat's state-of-the-art operator's cab is the product of a multi-year collaboration between Caterpillar, mining companies and operators. The cab provides industry-leading visibility with excellent line of sight supplemented by five optimally-mounted cameras and display screens. It provides enhanced safety through dual access/egress doors and an optimally-placed trainer seat with independent emergency stop. Additionally, the cab offers the smoothest, most comfortable ride available with an ergonomic, adjustable operator's seat with a fully pneumatic suspension system, low-effort joysticks and dual display screens optimized for operator comfort.

Custom Design and Aftermarket Support

The Cat dragline model selection process is grounded in analysis and collaboration with experienced application engineering professionals and a dragline optimization process to assist in determining the configuration optimally suited for a particular application. Additionally, mechanical and electrical upgrades, and component rebuilds are available to ensure productivity and reliability for the life of the machine. Caterpillar also offers machine relocation and field assembly services supported by experienced industry experts.

ELECTRIC ROPE SHOVELS

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INTRODUCTION

Electric rope shovels are one of the largest loading tools on the market. They run in a truck-shovel operation with mining trucks in order to move large quantities of material at a low cost per ton. Electric rope shovels are used to mine a variety of minerals including oil sands, coal overburden, copper, gold, iron ore, etc. They operate in extreme climates in tough surface mining applications all over the world. With over 130 years' experience in the rope shovel industry, and an active population of over 225 machines, Cat Electric Rope Shovels are the most efficient and cost effective loading tools available.

SAFETY AND SERVICEABILITY

Safety plays an integral role in everything we do at Caterpillar, from our factory floors to our service centers to our clients' mine sites. Caterpillar's commitment to safety is apparent in our product designs, which undergo risk assessments and are designed to meet strict codes and regulations.

Stairways, walkways, platforms and access points are incorporated throughout Cat Electric Rope Shovels to ensure safe and convenient access for maintenance personnel. With safety and serviceability enhancing features incorporated into the operator cab, structures, and electrical equipment, among others, Cat Electric Rope Shovels were designed with safety in mind.

FEATURES

AC IGBT Electrics

With over 40 years of AC experience and over 200 operating AC machines, Bucyrus led the industry in AC electric rope shovels. Caterpillar continues to carry on this proud tradition. Since its launch in 1981, the AC electric rope shovel has gained strong industry acceptance, offering:

- **Superior availability:** AC IGBT electric rope shovels have routinely demonstrated electrical availabilities of greater than 98%.
- **Greater reliability:** Fewer components including interchangeable inverters. IGBT systems require no fuses or circuit breakers.
- **Reduced maintenance:** No regular maintenance is required on IGBT power control modules. Motor maintenance on AC machines is reduced to greasing and replacing bearings every 30,000 hours.
- **Durability:** Cat IGBT is a mine quality system with rugged unitized construction to withstand harsh mining conditions.
- **Reduced inventory:** One IGBT part number is used in multiple locations allowing for decreased component inventory.

Major Structures

Cat major structures are designed for extended performance in harsh mining conditions. Structures are manufactured using cold-weather, impact-resistant, high-strength steel with select welds of full-penetration, profiled and ground type. All structural welds undergo visual inspection, with critical welds also receiving MT, UT or X-ray inspection. Large furnaces are used to stress-relieve entire weldments for reduced susceptibility to cracking. Interiors of finished structures are painted white to facilitate field inspection.

Robust Front End Design

The Cat crowd/retract system design presents many benefits over traditional rack-and-pinions systems, including:

- **Fast swing times:** The Cat deck-mounted, rather than boom-mounted, crowd machinery greatly reduces front end weight for reduced swing inertia and fast swing times.
- **Elimination of torsional loading:** The free-floating tubular handle design allows the Cat handle to rotate under uneven loading, transferring force into the ropes rather than into the boom.
- **Superior visibility:** With deck-mounted crowd machinery, Cat machines provide operators with a clear left-hand line of sight for higher visibility and enhanced safety.
- **Efficient digging:** Wide-set boom point sheaves stabilize the dipper as it engages the bank for easier and more efficient digging.

Cat Cab

Caterpillar's state-of-the-art operator's cab is the product of a multi-year collaboration between Caterpillar, mining companies and shovel operators. The cab provides industry-leading visibility supplemented by five optimally-mounted cameras and display screens. It provides enhanced safety through dual access/egress doors and an optimally-placed trainer seat with independent emergency stop. Additionally, the cab offers the smoothest, most comfortable ride available with an ergonomic, adjustable operator's seat with a fully pneumatic suspension system, low-effort joysticks and dual display screens optimized for operator comfort.

HydraCrowd™

HydraCrowd represents the first new crowd technology developed in the past 70 years. A hydraulic cylinder inside the tubular dipper handle maintains all the benefits of the Cat front end while eliminating the need for routine crowd/retract rope replacements. HydraCrowd cuts downtime by reducing the number of maintenance events needed to keep the machine operational. HydraCrowd is controlled with proven Cat IGBT technology and has complete diagnostic and troubleshooting information with step-by-step instructions.

LatchFree™

The LatchFree Dipper System is Caterpillar's solution to customers' number one cause of downtime — the traditional latch assembly. The LatchFree dipper eliminates the latch assembly, replacing it with a strong steel link mounted to the dipper back, away from material flow. The system enhances safety by reducing the number of maintenance events required to maintain the system and increases reliability by reducing unplanned downtime. The LatchFree Dipper System comes complete with a comprehensive training program to ensure customers achieve maximum system benefits.

FastFil™ Dipper Design

The unique FastFil dipper provides customers with faster and fuller dipper loads where the application permits. Its trapezoidal shape accommodates the natural configuration of the load, eliminating voids that occur with box-shaped dippers for improved fill factors. The trapezoidal shape also optimizes dipper size and weight for improved maneuverability and faster swing times. Additionally, an adjustable pitch brace allows rake angle changes to improve bank penetration, eliminate bulldozing, and improve productivity. Combined, the FastFil features maximize the fill factor.

Operator Assist — Enhanced Motion Control

The operator assist — enhanced motion control system eliminates or reduces the occurrence of crowd over-speeds, crowd impacts, hoist stalls, boom jacking, and swinging while engaged in the bank. This is accomplished by intelligently controlling how motions are allowed to operate under certain conditions. The reduced system stress and equal or better cycle times improve shovel performance and cost per ton. This software is standard on all 7495 Series electric rope shovels.

Product Link Elite

This sytem boosts connectivity and increases the availability of data provided by the shovel. The onboard hardware enables the shovel to collect and transmit information into locally hosted or cloud-hosted applications such as Cat MineStar™ Solutions Health Office, Health Equipment Insights, Equipment Care Advisor or Vision Link.

PTM Payload

Loadcell technology directly measures padlock pin force and dipper accelerations to determine payload, thereby providing real time feedback. It is an improvement over traditional systems, which are quasi-static methods that rely on averaging to estimate dipper force and inertial effects. PTM Payload can calculate material moved, monitor overloading of trucks, and empower operators to study their loading efficiency and shift performance.

Cycle Segmentation

Standard on new machines, this feature detects the machine operation mode and classifies the severity of those operations (dig, swing to dump, dump, and return to dig). It creates application profiles and composite work cycles based on machine data. The data is stored onboard to provide key performance indicators and operator feedback in real time. It is also transmitted offboard through Product Link Elite for back office applications.

Integrated Technology Platform

This hardware provides the foundation to seamlessly integrate software solutions as well as interface with the MineStar™ product portfolio. It provides a multi-generational building block approach to mature and launch machine-level technology solutions digitally, eliminating future work stoppage for additional hardware installation. It enables deployment of onboard machine technologies that support higher productivity, improved reliability, and increased cyber security. The Technology Controls network provides simple maintenance, troubleshooting and feature upgrades.

Integral Fast Propel Transfer Switch

The Cat integral fast propel transfer switch is a more responsive system that improves mode transfer as much as 75% — delivering more production hours and reducing service and costs due to downtime.

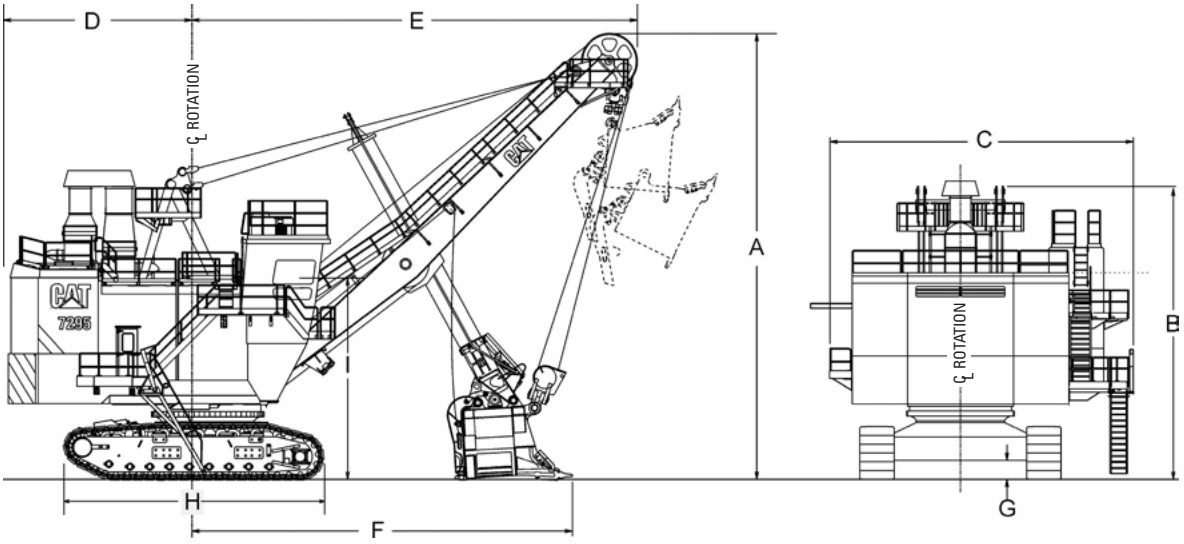
MineStar™ Health

MineStar™ Health is an industry-leading technology offering that helps maximize equipment availability and reliability. MineStar™ Health keeps you connected to your machines so you can head off small problems while they're still small, run machines as efficiently as possible for as long as possible, and keep unplanned downtime to a minimum.

MineStar™ Health products and services enable you to collect and transmit equipment data, monitor critical machine parameters, obtain real-time alerts, analyze operational trends and patterns, predict failures and receive repair recommendations — assisting you with proactive maintenance services and predictive equipment analysis.

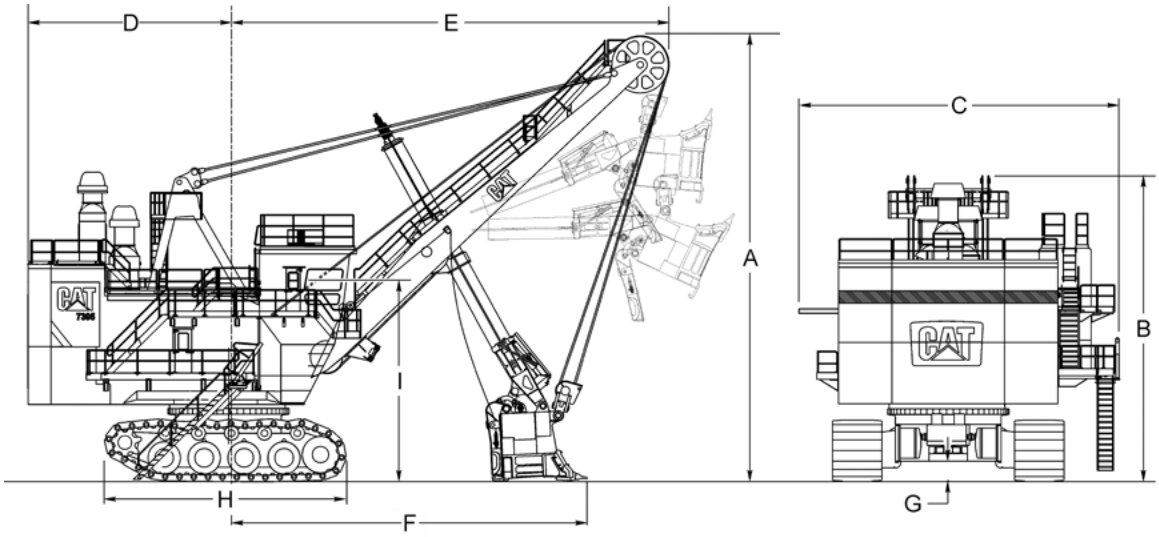
MineStar™ Terrain

MineStar™ Terrain for loading is a machine guidance system that delivers real-time data in the cab—on everything from bucket positioning and bench height to ore bodies and volume of material cut and filled. That enables every operator to maximize machine efficiency, leading to better material management, more accurate grades and less rework and wear.



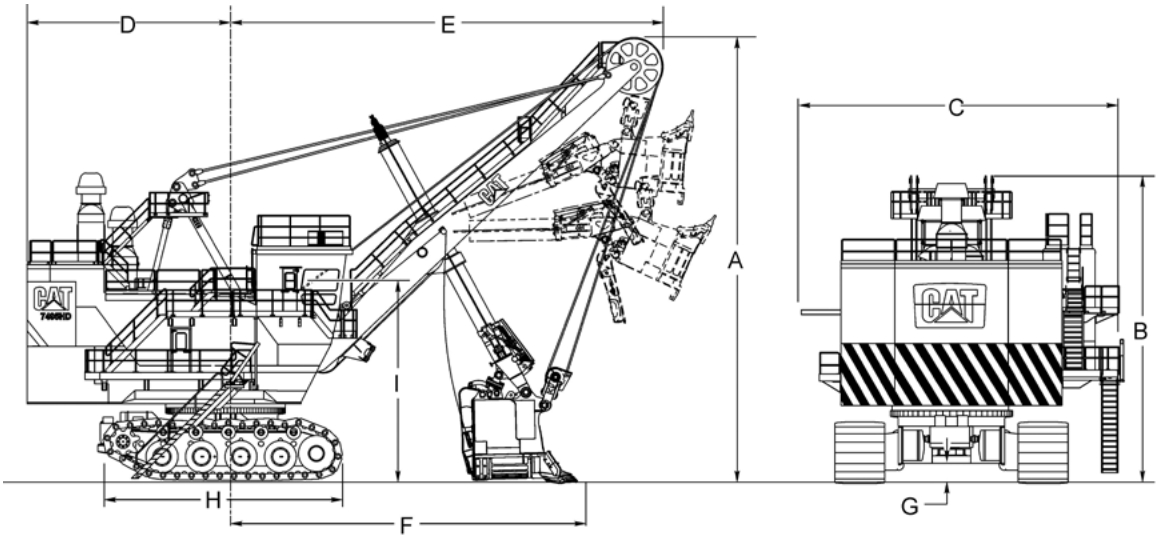
MODEL		7295	
Boom Length		18.00 m	**59'0"**
A	Height	18.15 m	**59'7"**
B	A-Frame Height	11.94 m	**39'2"**
C	Overall Width	12.39 m	**40'8"**
D	Tail Swing Radius	7.80 m	**25"8"**
E	Clearance Radius	17.48 m	**57'4"**
F	Radius of Level Floor*	14.90 m	**48'11"**
G	Minimum Ground Clearance	0.78 m	**2'6"**
H	Track Length	10.26 m	**33'8"**
I	Operator Eye Level	8.20 m	**26'11"**

*Dimensions based on 19 m^3 **(25 yd^3)** dipper.
Information subject to change.
All dimensions are approximate and will vary depending on dipper size.



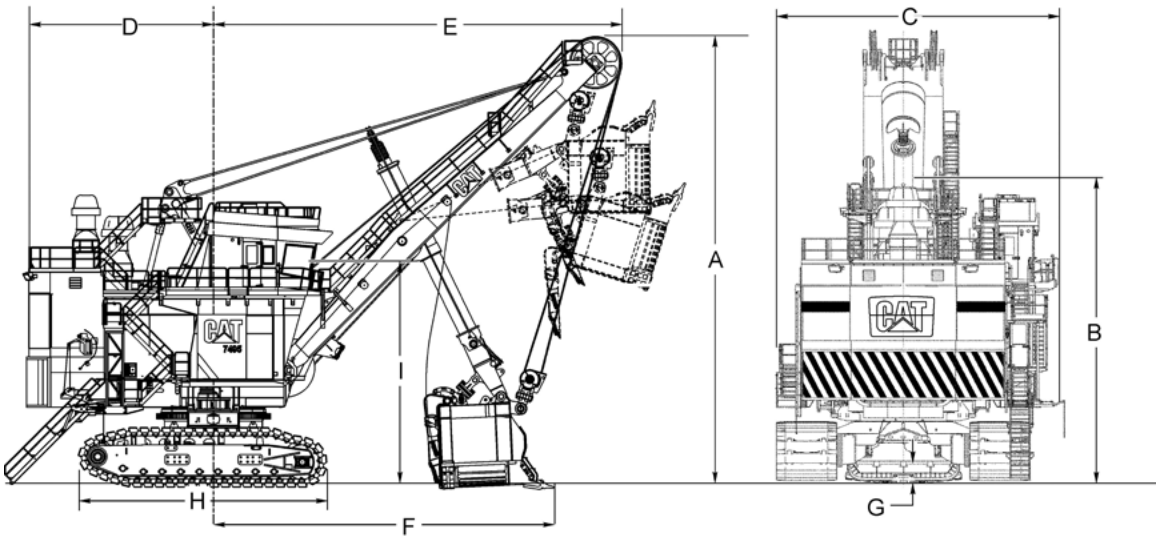
MODEL		7395	
Boom Length		19.51 m	**64'0"**
A	Height	19.50 m	**63'11"**
B	A-Frame Height	13.30 m	**43'8"**
C	Overall Width	13.01 m	**42'8"**
D	Tail Swing Radius	9.30 m	**30'8"**
E	Clearance Radius	19.05 m	**62'6"**
F	Radius of Level Floor*	16.00 m	**52'4"**
G	Minimum Ground Clearance	0.86 m	**2'10"**
H	Track Length	10.41 m	**34'2"**
I	Operator Eye Level	8.64 m	**28'4"**

*Dimensions based on 24 m^3 **(32 yd^3)** dipper.
Information subject to change.
All dimensions are approximate and will vary depending on dipper size.



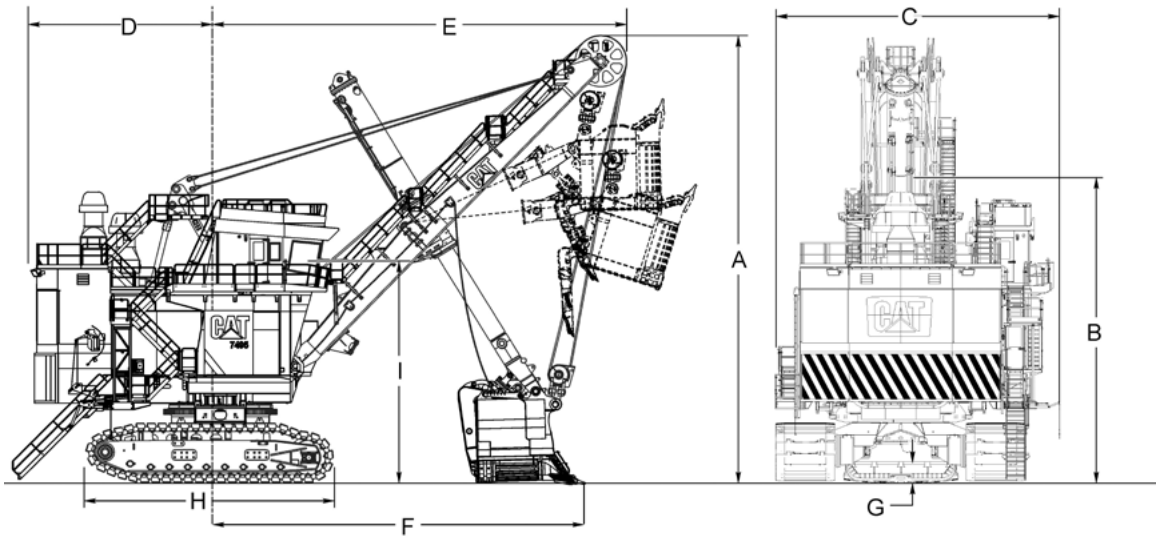
MODEL		7495 HD
Boom Length	20.40 m	**67'0"**
A Height	20.10 m	**65'10"**
B A-Frame Height	13.30 m	**43'8"**
C Overall Width	13.01 m	**42'8"**
D Tail Swing Radius	9.30 m	**30'8"**
E Clearance Radius	19.74 m	**64'9"**
F Radius of Level Floor*	15.70 m	**51'8"**
G Minimum Ground Clearance	0.86 m	**2'10"**
H Track Length	10.41 m	**34'2"**
I Operator Eye Level	8.64 m	**28'4"**

*Dimensions based on 32 m^3 **(42 yd^3)** dipper.
Information subject to change.
All dimensions are approximate and will vary depending on dipper size.



MODEL		7495	
Boom Length		20.40 m	**67'0"**
A	Height	20.87 m	**68'6"**
B	A-Frame Height	14.00 m	**46'0"**
C	Overall Width	13.11 m	**43'0"**
D	Tail Swing Radius	9.30 m	**30'8"**
E	Clearance Radius	19.65 m	**64'5"**
F	Radius of Level Floor*	17.10 m	**56'1"**
G	Minimum Ground Clearance	0.90 m	**3'0"**
H	Track Length	11.43 m	**37'6"**
I	Operator Eye Level	10.61 m	**34'10"**

*Dimensions based on 56 m^3 **(73 yd^3)** dipper.
Information subject to change.
All dimensions are approximate and will vary depending on dipper size.



MODEL		7495 HydraCrowd	
Boom Length		20.40 m	**67'0"**
A	Height	20.87 m	**68'6"**
B	A-Frame Height	14.00 m	**46'0"**
C	Overall Width	13.11 m	**43'0"**
D	Tail Swing Radius	9.30 m	**30'8"**
E	Clearance Radius	19.65 m	**64'5"**
F	Radius of Level Floor*	17.10 m	**56'1"**
G	Minimum Ground Clearance	0.90 m	**3'0"**
H	Track Length	11.43 m	**37'6"**
I	Operator Eye Level	10.61 m	**34'10"**

*Dimensions based on 56 m^3 **(73 yd^3)** dipper.
Information subject to change.
All dimensions are approximate and will vary depending on dipper size.